

Basic Prob. Facts LOTUS: $E[g(X)] = \sum_x g(x) P(X=x)$ Tower: $E[E[X|Y]] = E[X]$

Expectation: $E[X] = \sum_x x P(X=x)$ Conditional Prob: $P(X=x|Y=y)$, $\sum_x P(X=x|Y=y) = 1$

Variance: $Var(X) = E[(X-E[X])^2] = E[X^2] - E[X]^2$ Total exp: $E[X] = \sum_y P(Y=y) E[X|Y=y]$

$Var(\sum_i X_i) = \sum_i \sum_j Cov(X_i, X_j) = \sum_i Var(X_i) + 2 \sum_{i < j} Cov(X_i, X_j)$ MGF: $M_X(s) = E[e^{sx}] = \sum_x e^{sx} P(X=x)$

Covariance: $Cov(X, Y) = E[(X-E[X])(Y-E[Y])] = E[XY] - E[X]E[Y]$ $\frac{d^n}{ds^n} M_X(s)|_{s=0} = E[X^n]$

$Cov(X, aY+b) = a Cov(X, Y)$, $Cov(X, Y+Z) = Cov(X, Y) + Cov(X, Z)$ $E[e^{sx}] = E[\sum_x \frac{e^{sx}}{h!} X^h] \leq \sum_x \frac{e^{sx}}{h!} X^h$

Concentration Inequalities

Markov: $P(X \geq a) \leq \frac{E[X]}{a}$

Chebyshev: $P(|X-E[X]| \geq \epsilon) \leq \frac{Var(X)}{\epsilon^2}$ Convergence almost sure: $(X_n, n \geq 1)$ converges a.s. to X_∞ if $P(A) = 1$ where $A = \{ \omega: X_n(\omega) \rightarrow X_\infty(\omega) \text{ as } n \rightarrow \infty \}$

Bounded Convergence Thm.

$X_n \xrightarrow{a.s.} X_\infty$ as $n \rightarrow \infty$ and $\exists C$ s.t. $|X_n(\omega)| \leq C < \infty$ for $n \geq 0$ and $\omega \in \Omega$. Then $E[X_n] \rightarrow E[X_\infty]$ as $n \rightarrow \infty$

Dominated Convergence Thm.

$X_n \xrightarrow{a.s.} X_\infty$ a.s. as $n \rightarrow \infty$ and $\exists$ r.v. Y s.t. $E[Y] < \infty$ and $|X_n(\omega)| \leq Y(\omega)$ for $n \geq 0, \omega \in \Omega$.

Then, $E[X_n] \rightarrow E[X_\infty]$ as $n \rightarrow \infty$

CLT: $(X_n)_{n=1}^\infty$ iid with mean μ , variance σ^2 . SLN : $(X_n)_{n=1}^\infty$ iid, X_i 's bounded. $\bar{X}_n = \frac{1}{n} \sum_{i=1}^n X_i \xrightarrow{a.s.} E[X_1]$

$\hookrightarrow \frac{\bar{X}_n - \mu}{\sigma/\sqrt{n}} \xrightarrow{indist} N(0,1)$ $WLLN$: iid $Z_1, \dots, Z_n$. $\bar{Z}_n \xrightarrow{P} E[Z_1]$

Markov Chains

Markov Property: $P(X_{n+1} \in B | X_0, \dots, X_n) = P(X_{n+1} \in B | X_n) = P(X_1 \in B | X_0) = P(X_n, X_{n+1})$

Stationary distribution: chain will have the same dist for all time: $\pi P = \pi$ (solve s.t. $\pi e = 1$) $P = \lim_{n \rightarrow \infty} P^n$

$P(X, B) = \sum_{b \in B} P(X, b)$ or $P(X, B) = \int P(X, b) db$ state is transient iff $\sum_{n=0}^\infty P^n(X, y) < \infty$

Hitting time $T = \inf \{ n \geq 0: X_n \in C \}$ (first time we hit C) Recurrent otherwise

n -step transition probabilities: $\gcd \{ n \geq 1: P^n(x, x) > 0 \} = p \Rightarrow x \in S$ is periodic of period p . If the period is 1, the state is aperiodic.

For an irreducible positive recurrent aperiodic MC, $P^n(x, y) \rightarrow \pi(y)$ as $n \rightarrow \infty$ where $\pi(y)$ is the stationary dist at state y .

When you condition an MC on the outcome of an event, it's an MC. $\tilde{P}(x, y) = \frac{\pi(y) P(x, y)}{\pi(x)}$

Reversing an MC: $X_i = X_j$ also an MC with transition matrix $\tilde{P}(x, y) = \frac{\pi(y) P(y, x)}{\pi(x)}$

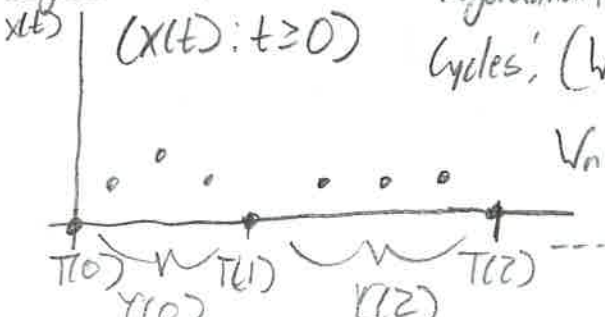
Global balance: $\pi = \pi P$ Detailed balance: $\pi(x) P(x, y) = \pi(y) P(y, x) \forall x, y \in S$

Martingale SLN

$\frac{1}{n} \sum_{i=1}^n P_i \rightarrow 0$ a.s. (if $\sum P_i < \infty$)
Poisson's Eqn
 X (finite state irreducible)
 $(P-I)f = h$

potential or accumulated deviation from the mean
 cost/reward fraction

Regenerative Processes



Strong Law for Regenerative Processes:

If $\int_0^{T(1)} f(X(s)) ds < \infty$ a.s.
 $E[\int_0^{T(1)} f(X(s)) ds]$ and $E[Y_1]$ are not simultaneously infinite
 Then $\frac{1}{t} \int_0^t f(X(s)) ds \rightarrow \frac{E[Y_1(f)]}{E[Y_1]}$ a.s.
 where $Y_i(f) = \int_{T(i-1)}^{T(i)} f(X(s)) ds$

Harris Chains (recurrence without strong $\pi = \pi P$)

MC $(X_n: n \geq 0)$ is Harris recurrent if $\exists A \subseteq S, \lambda > 0, m \geq 1$ and a distribution ψ on S s.t.
 i) $P_x(T_A < \infty) = 1, x \in S$, where $T_A = \inf\{n \geq 0: X_n \in A\}$
 ii) $P_x(X_m \in \cdot) \geq \lambda \psi(\cdot), x \in A$
 If X is discrete, X is Harris recurrent iff $\exists z \in S$ s.t. $P_x(\tau(z) < \infty) = 1$ for $x \in S$, where $\tau(z) = \inf\{n \geq 1: X_n = z\}$
 For Harris recurrent MC and non-negative f , $\frac{1}{n} \sum_{i=0}^{n-1} f(X_i) \rightarrow \int \pi(dx) f(x)$ a.s. as $n \rightarrow \infty$

Martingale CLT

$(M_n: n \geq 0)$ square integrable
 $\frac{1}{n} \sum_{i=1}^n [M_i](n) \xrightarrow{p} \sigma^2$ as $n \rightarrow \infty$
 $[M](n) = M^2(n) - \sum_{i=1}^n P_i^2$
 $\frac{1}{n} \sum_{i=1}^n \langle M \rangle(i) \xrightarrow{p} \sigma^2$ as $n \rightarrow \infty$
 $\langle M \rangle(n) = \sum_{i=1}^n E[P_i^2 | \mathcal{F}_{i-1}]$
 Then $\frac{1}{\sqrt{n}} \sum_{i=1}^n P_i \rightarrow N(0, \sigma^2)$

Martingale Convergence Thm

$(M_n: n \geq 0)$ adapted to $(\mathcal{F}_n: n \geq 0)$
 If $\sup_{n \geq 0} E[M_n^2] < \infty, \exists$ square integrable r.v. M_∞ s.t.
 $M_n \rightarrow M_\infty$ a.s. as $n \rightarrow \infty$
Supermartingale Convergence Thm
 $(M_n: n \geq 0)$ super MC, non-negative
 Then, $\exists$ finite-valued r.v. M_∞ s.t. $M_n \rightarrow M_\infty$ a.s. as $n \rightarrow \infty$

Data-Driven Markov Modeling

Goal: learn parameters θ to learn the transition probabilities, i.e.
 $\hat{\theta} = \arg \max_{\theta} d(\theta) = \sum_{i=1}^n \log P(\theta, X_i, X_{i+1})$
 $\Rightarrow M_n = \sum_{i=0}^{n-1} \frac{\nabla P(\theta, X_i, X_{i+1})}{P(\theta, X_i, X_{i+1})}$ is MC
 And $\sqrt{n}(\hat{\theta}_n - \theta^*) H_n = \frac{-M_n}{n} \rightarrow N(0, S)$
 $S = E[P'(\theta^*)^2]$

Regeneration times: $(T(n): n \geq 0)$

Cycles: $(V_n: n \geq 0)$
 $V_n(t) = \sum_{i=0}^{n-1} X(T(i-1) + t) \leq t \leq T_n$
 $\Delta t \geq T_n$
 This regenerative if
 • $V_0, V_1, \dots$ are independent
 • $V_1, V_2, \dots$ are iid
 When $T(0) = 0$, this is a renewal counting process.

CLT for Regenerative Processes

$X = (X(t): t \geq 0)$ S-valued regenerative process with regeneration times $(T(n): n \geq 0)$
 Sps. $f: S \rightarrow \mathbb{R}$ satisfies $Y_0(f) < \infty$ a.s. and $E[Y_1(f)^2 + Y_1] < \infty$. Then
 $\sqrt{t} \left(\frac{1}{t} \int_0^t f(X(s)) ds - \alpha \right) \Rightarrow N(0, 1)$
 where $\alpha = \frac{E[Y_1(f)]}{E[Y_1]}, \sigma^2 = \frac{E[Z_1^2]}{E[Y_1]}$
 and $Z_1 = Y_1(f) - \alpha Y_1$

$\frac{N(t)}{t} \rightarrow \frac{1}{E[Y_1]}$ a.s. as $t \rightarrow \infty$

Elementary renewal thm:

$\frac{E[N(t)]}{t} \rightarrow \frac{1}{E[Y_1]}$ as $t \rightarrow \infty$
Renewal A recurrent state $z \in S$ is said to be positive recurrent if $E[\tau(z)] < \infty$ and null recurrent otherwise
 Let $X = (X_n: n \geq 0)$ be MC.
 If X is null recurrent and $f: S \rightarrow \mathbb{R}_+$ s.t. $E[Y_1(f)] < \infty$, then $\frac{1}{n} \sum_{i=0}^{n-1} f(X_i) \rightarrow 0$ a.s. as $n \rightarrow \infty$. Also, if $E[Y_1(f)] < \infty$, then $\frac{1}{n} \sum_{i=0}^{n-1} E_X[f(X_i)] \rightarrow 0$

Thm. 11.2.1. Sps. $\exists$ AES $\lambda > 0, m \geq 1$, a distribution $\psi, \psi > 0$, and $g: S \rightarrow [0, \infty)$ s.t.

- i) $E_x[g(X_1)] \leq g(x) - \epsilon, x \in A^c$
- ii) $\sup_{x \in A} E_x[g(X_1)] < \infty$
- iii) $P_x(X_m \in \cdot) \geq \lambda \psi(\cdot), x \in A$

then, we have a Harris chain.

Also, g is the Lyapunov function. (common candidates for g include $\|x\|^p, \exp(\|x\|^p)$, and $(\log(1 + \|x\|))^p$)

Maximal Inequality

$$P(\max_{1 \leq i \leq n} M_i > x) \leq \frac{E[M_n]}{x}$$

Thm. 11.2.1. Sps. $\exists$ AES $\lambda > 0, m \geq 1$, a distribution $\psi, \psi > 0$, and $g: S \rightarrow [0, \infty)$ s.t.

- a. If X is null recurrent and $f: S \rightarrow \mathbb{R}_+$ s.t. $E[Y_1(f)] < \infty$, then $\frac{1}{n} \sum_{i=0}^{n-1} f(X_i) \rightarrow 0$ a.s. as $n \rightarrow \infty$. Also, if $E[Y_1(f)] < \infty$, then $\frac{1}{n} \sum_{i=0}^{n-1} E_X[f(X_i)] \rightarrow 0$
- b. If X is positive recurrent iff $\exists$ a prob. soln. to $\pi = \pi P$
- c. If X is positive recurrent, then for each $f: S \rightarrow \mathbb{R}_+, \frac{1}{n} \sum_{i=0}^{n-1} f(X_i) \rightarrow \int \pi(dy) f(y)$ a.s. as $n \rightarrow \infty$ and $\frac{1}{n} \sum_{i=0}^{n-1} E_X[f(X_i)] \rightarrow \int \pi(dy) f(y)$ as $n \rightarrow \infty$

Also, $\pi(z) = \frac{1}{E_2[\tau(z)]}$

Constructing MC

$X = (X_n: n \geq 0)$ is a Markov chain with $X_n = (\phi_n \circ \phi_{n-1} \circ \dots \circ \phi_1)(x)$ ϕ_i 's iid
 $E[d(\phi(x), \phi(y))] \leq C d(x, y) < 1$